

FFNN Hyperparameter Grid Version

August 18, 2025

```
[15]: # This is an example of the code used to grid search through the FFNN...
      # it takes a very long time to execute the computations and is here as an
      ↪example but not used in final code
```

```
[16]: # Import necessary libraries
import pandas as pd
import numpy as np
from scipy.sparse import csr_matrix, hstack
from sklearn.preprocessing import OneHotEncoder, MinMaxScaler
from sklearn.model_selection import train_test_split
import tensorflow as tf
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense
from tensorflow.keras.layers import LeakyReLU
from tensorflow.keras.layers import ReLU
from tensorflow.keras.optimizers import Adam, RMSprop, SGD
from tensorflow.keras.callbacks import EarlyStopping# Added for tuning
import matplotlib.pyplot as plt
import pyhere
```

```
[17]: train_path = pyhere.here('Data', 'train_data.xlsx')
test_path = pyhere.here('Data', 'test_data.xlsx')
train_data = pd.read_excel(train_path)
test_data = pd.read_excel(test_path)
# Use dropna() to remove any rows with missing values
train_data = train_data.dropna()
test_data = test_data.dropna()
print(train_data.head())
```

	CountryName	CountryCode	year	HDI_Index	\
0	Antigua and Barbuda	ATG	2022	0.848	
9	Antigua and Barbuda	ATG	2008	0.830	
10	Antigua and Barbuda	ATG	2009	0.827	
11	Antigua and Barbuda	ATG	2010	0.828	
12	Antigua and Barbuda	ATG	2011	0.828	

	HDI_Category	FoodIndex	ElectricAccess	PopDensity	\
0	Very high human development	89.95	100.0	211.000000	

9	Very high human development	145.98	100.0	188.436364
10	Very high human development	141.40	98.5	191.302273
11	Very high human development	102.49	98.7	193.909091
12	Very high human development	104.99	94.6	196.209091

	PopInSlums	WaterStress	...	IncomeGroup	YearlyChgInternet	\
0	2.646340	8.461538	...	High income	0.997844	
9	2.646317	9.230769	...	High income	3.164091	
10	2.646311	9.038462	...	High income	3.164091	
11	2.646300	8.846154	...	High income	3.831893	
12	2.646300	8.653846	...	High income	3.831893	

	YearlyChgHDI	Lag1_InternetUsersPct	Lag2_InternetUsersPct	\
0	0.005	8.247411	7.147796	
9	0.003	5.957120	5.079504	
10	-0.003	6.242918	5.336653	
11	0.001	6.508026	5.571669	
12	0.000	6.815164	5.788459	

	Lag1_YearlyChgInternet	Lag2_YearlyChgInternet	\
0	0.993070	0.908060	
9	3.128215	2.432146	
10	3.128215	3.116179	
11	3.128215	3.116179	
12	3.782704	3.116179	

	Cumulative3yrChg_InternetUsersPct	GovernanceScore	HealthSpend
0	3.2	0.180504	853.834335
9	11.0	0.904398	523.176632
10	12.0	0.929469	472.044125
11	13.0	0.914123	573.175838
12	14.0	0.898586	562.880747

[5 rows x 30 columns]

```
[18]: train_target = train_data['InternetUsersPct']
      test_target = test_data['InternetUsersPct']

      train_predictors = train_data.drop('InternetUsersPct', axis=1)
      test_predictors = test_data.drop('InternetUsersPct', axis=1)

[27]: # Combine the predictor data frames to ensure consistent encoding
      combined_predictors = pd.concat([train_predictors, test_predictors],
      ↪ ignore_index=True)

      # Identify numerical and categorical columns
      numerical_cols = combined_predictors.select_dtypes(include=np.number).columns
```

```

categorical_cols = combined_predictors.select_dtypes(include='object').columns

# Initialize MinMaxScaler
scaler = MinMaxScaler()

# Fit the scaler on the combined numerical data to learn the min/max values
combined_predictors[numerical_cols] = scaler.
    ↪fit_transform(combined_predictors[numerical_cols])

# Initialize OneHotEncoder to create sparse output and handle unknown categories
ohe = OneHotEncoder(sparse_output=False, handle_unknown='ignore')

# Fit the encoder on the combined data to learn all possible categories
combined_categorical_encoded = ohe.
    ↪fit_transform(combined_predictors[categorical_cols])

# Combine the encoded categorical data with the numerical data
combined_features = np.concatenate([combined_predictors[numerical_cols].values,
                                     combined_categorical_encoded], axis=1)

# Split the combined feature matrix back into training and test sets
train_features = combined_features[:len(train_predictors)]
test_features = combined_features[len(train_predictors):]

# Features split
train_features, test_features, train_target, test_target = train_test_split(
    combined_features, train_target, test_size=0.2, random_state=42
)

```

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[28]: # Hyperparameter tuning with grid search
hyper_grid = {
    'epochs': [1000, 2000],
    'batch_size': [16, 32, 64, 128],
    'learning_rate': [0.0001, 0.001, 0.01, 0.1]
}
results = []

```

```

[29]: # Hyperparameter tuning loop
for epochs in hyper_grid['epochs']:
    for batch_size in hyper_grid['batch_size']:
        for lr in hyper_grid['learning_rate']:
            print(f"\nTraining with epochs={epochs}, batch_size={batch_size},
    ↪lr={lr}")

            model = Sequential()
            model.add(Dense(64, input_dim=train_features.shape[1]))
            model.add(ReLU())

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model.add(Dense(32))
model.add(ReLU())
model.add(Dense(1, activation='linear'))

optimizer = RMSprop(learning_rate=lr)
model.compile(optimizer=optimizer, loss='mean_absolute_error',
↳metrics=['mae'])

early_stop = EarlyStopping(monitor='val_loss', patience=50,
↳verbose=0, restore_best_weights=True)

history = model.fit(
    train_features, train_target,
    epochs=epochs,
    batch_size=batch_size,
    validation_split=0.2,
    callbacks=[early_stop],
    verbose=0
)

# Evaluate safely
if test_features.shape[0] == 0 or test_target.shape[0] == 0:
    print("Skipping evaluation: Test data is empty.")
    loss, mae = None, None
else:
    eval_results = model.evaluate(test_features, test_target,
↳verbose=0)

    if isinstance(eval_results, (list, tuple)):
        if len(eval_results) >= 2:
            loss, mae = eval_results[0], eval_results[1]
        elif len(eval_results) == 1:
            loss, mae = eval_results[0], None
        elif len(eval_results) == 0:
            print("evaluate() returned empty list. Setting loss/mae
↳= None.")
            loss, mae = None, None
        else:
            loss, mae = eval_results, None

    results.append({
        'epochs': epochs,
        'batch_size': batch_size,
        'learning_rate': lr,
        'loss': loss,
        'mae': mae
    })

```

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# Put results in DataFrame
results_df = pd.DataFrame(results)
results_df = results_df.dropna(subset=['loss'])

if not results_df.empty:
    print("\nAll Hyperparameter Results:")
    print(results_df)

    best_params = results_df.loc[results_df['loss'].idxmin()]
    print("\nBest Hyperparameters Found:")
    print(best_params)

    # Train the Final Model with Best Parameters
    print("\nTraining final model with best parameters...")
    final_model = Sequential()
    final_model.add(Dense(256, input_dim=train_features.shape[1],
↪activation='relu'))
    final_model.add(Dense(128, activation='relu'))
    final_model.add(Dense(64, activation='relu'))
    final_model.add(Dense(1, activation='linear'))

    final_optimizer = RMSprop(learning_rate=best_params['learning_rate'])
    final_model.compile(optimizer=final_optimizer, loss='mean_absolute_error',
↪metrics=['mae'])

    early_stop_final = EarlyStopping(monitor='val_loss', patience=50,
↪verbose=1, restore_best_weights=True)

    final_history = final_model.fit(
        train_features, train_target,
        epochs=1000,
        batch_size=int(best_params['batch_size']),
        validation_data=(test_features, test_target),
        callbacks=[early_stop_final],
        verbose=1
    )

    # Final Evaluation
    if test_features.shape[0] == 0 or test_target.shape[0] == 0:
        print("Skipping final evaluation: Test data is empty.")
        loss_final, mae_final = None, None
    else:
        eval_results_final = final_model.evaluate(test_features, test_target,
↪verbose=0)

        if isinstance(eval_results_final, (list, tuple)):

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        if len(eval_results_final) >= 2:
            loss_final, mae_final = eval_results_final[0],
↪eval_results_final[1]
        elif len(eval_results_final) == 1:
            loss_final, mae_final = eval_results_final[0], None
        elif len(eval_results_final) == 0:
            print("evaluate() returned empty list for final model.")
            loss_final, mae_final = None, None
    else:
        loss_final, mae_final = eval_results_final, None

    if loss_final is not None:
        print(f"\nFinal Model Loss on Test Data: {loss_final:.4f}")
    if mae_final is not None:
        print(f"Final Model MAE on Test Data: {mae_final:.4f}")

    # Predictions and Plots
    predictions = final_model.predict(test_features)
    if not np.any(np.isnan(predictions)):
        predictions = predictions.flatten()
        final_results_df = pd.DataFrame({'Actual': test_target, 'Predicted':
↪ predictions})

    # Scatter plot
    plt.figure(figsize=(10, 6))
    plt.scatter(final_results_df['Actual'],
↪final_results_df['Predicted'], alpha=0.6)
    plt.plot([min(final_results_df['Actual']),
↪max(final_results_df['Actual'])],
             [min(final_results_df['Actual']),
↪max(final_results_df['Actual'])],
             color='red', linestyle='--', lw=2)
    plt.title('Actual vs. Predicted Values')
    plt.xlabel('Actual Internet Users Percentage')
    plt.ylabel('Predicted Internet Users Percentage')
    plt.show()

    # Residual plot
    final_results_df['Residuals'] = final_results_df['Actual'] -
↪final_results_df['Predicted']
    plt.figure(figsize=(10, 6))
    plt.hist(final_results_df['Residuals'], bins=30, edgecolor='black',
↪alpha=0.7)
    plt.axvline(x=0, color='red', linestyle='--', lw=2)
    plt.title('Residuals Distribution')
    plt.xlabel('Residuals (Actual - Predicted)')

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plt.ylabel('Frequency')
plt.show()

# Training & validation MAE
mae_history = final_history.history['mae']
val_mae_history = final_history.history['val_mae']
epochs_range = range(1, len(mae_history) + 1)
plt.figure(figsize=(10, 6))
plt.plot(epochs_range, mae_history, 'b', label='Training MAE')
plt.plot(epochs_range, val_mae_history, 'r', label='Validation MAE')
plt.title('Training and Validation MAE over Epochs')
plt.xlabel('Epochs')
plt.ylabel('MAE')
plt.legend()
plt.show()
else:
    print("Predictions contain invalid values (NaN or infinity). Cannot_
    plot.")
else:
    print("No valid results found. Cannot train final model.")

```

Training with epochs=1000, batch_size=16, lr=0.0001

C:\Users\barre\my_new_env2\Lib\site-packages\keras\src\layers\core\dense.py:92:
 UserWarning: Do not pass an `input\_shape`/`input\_dim` argument to a layer. When
 using Sequential models, prefer using an `Input(shape)` object as the first
 layer in the model instead.
 super().__init__(activity_regularizer=activity_regularizer, **kwargs)

Training with epochs=1000, batch_size=16, lr=0.001

C:\Users\barre\my_new_env2\Lib\site-packages\keras\src\layers\core\dense.py:92:
 UserWarning: Do not pass an `input\_shape`/`input\_dim` argument to a layer. When
 using Sequential models, prefer using an `Input(shape)` object as the first
 layer in the model instead.
 super().__init__(activity_regularizer=activity_regularizer, **kwargs)

Training with epochs=1000, batch_size=16, lr=0.01

C:\Users\barre\my_new_env2\Lib\site-packages\keras\src\layers\core\dense.py:92:
 UserWarning: Do not pass an `input\_shape`/`input\_dim` argument to a layer. When
 using Sequential models, prefer using an `Input(shape)` object as the first
 layer in the model instead.
 super().__init__(activity_regularizer=activity_regularizer, **kwargs)

Training with epochs=1000, batch_size=16, lr=0.1

C:\Users\barre\my_new_env2\Lib\site-packages\keras\src\layers\core\dense.py:92:
UserWarning: Do not pass an `input\_shape`/`input\_dim` argument to a layer. When
using Sequential models, prefer using an `Input(shape)` object as the first
layer in the model instead.

```
super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

Training with epochs=1000, batch_size=32, lr=0.0001

C:\Users\barre\my_new_env2\Lib\site-packages\keras\src\layers\core\dense.py:92:
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super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

Training with epochs=1000, batch_size=32, lr=0.001

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```
super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

Training with epochs=1000, batch_size=32, lr=0.01

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layer in the model instead.

```
super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

Training with epochs=1000, batch_size=32, lr=0.1

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```
super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

Training with epochs=1000, batch_size=64, lr=0.0001

C:\Users\barre\my_new_env2\Lib\site-packages\keras\src\layers\core\dense.py:92:
UserWarning: Do not pass an `input\_shape`/`input\_dim` argument to a layer. When
using Sequential models, prefer using an `Input(shape)` object as the first
layer in the model instead.

```
super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

Training with epochs=1000, batch_size=64, lr=0.001

C:\Users\barre\my_new_env2\Lib\site-packages\keras\src\layers\core\dense.py:92:
UserWarning: Do not pass an `input\_shape`/`input\_dim` argument to a layer. When
using Sequential models, prefer using an `Input(shape)` object as the first
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```
super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

Training with epochs=1000, batch_size=64, lr=0.01

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super().__init__(activity_regularizer=activity_regularizer, **kwargs)
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Training with epochs=1000, batch_size=64, lr=0.1

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```
super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

Training with epochs=1000, batch_size=128, lr=0.0001

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Training with epochs=1000, batch_size=128, lr=0.001

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Training with epochs=1000, batch_size=128, lr=0.01

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Training with epochs=1000, batch_size=128, lr=0.1

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Training with epochs=2000, batch_size=16, lr=0.0001

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Training with epochs=2000, batch_size=32, lr=0.001

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```
Training with epochs=2000, batch_size=32, lr=0.1
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```

```
    super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

```
Training with epochs=2000, batch_size=64, lr=0.0001
```

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C:\Users\barre\my_new_env2\Lib\site-packages\keras\src\layers\core\dense.py:92:
UserWarning: Do not pass an `input_shape`/`input_dim` argument to a layer. When
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```
Training with epochs=2000, batch_size=64, lr=0.001
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Training with epochs=2000, batch_size=64, lr=0.01
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```

```
super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

Training with epochs=2000, batch_size=128, lr=0.0001

```
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UserWarning: Do not pass an `input_shape`/`input_dim` argument to a layer. When
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```

```
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```

Training with epochs=2000, batch_size=128, lr=0.001

```
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```

Training with epochs=2000, batch_size=128, lr=0.01

```
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super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

Training with epochs=2000, batch_size=128, lr=0.1

```
C:\Users\barre\my_new_env2\Lib\site-packages\keras\src\layers\core\dense.py:92:
UserWarning: Do not pass an `input_shape`/`input_dim` argument to a layer. When
using Sequential models, prefer using an `Input(shape)` object as the first
layer in the model instead.
```

```
super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

All Hyperparameter Results:

	epochs	batch_size	learning_rate	loss	mae
0	1000	16	0.0001	0.116306	0.116306
1	1000	16	0.0010	0.120173	0.120173
2	1000	16	0.0100	0.143841	0.143841
3	1000	16	0.1000	0.412924	0.412924
4	1000	32	0.0001	0.107589	0.107589
5	1000	32	0.0010	0.157023	0.157023
6	1000	32	0.0100	0.194848	0.194848

7	1000	32	0.1000	0.339577	0.339577
8	1000	64	0.0001	0.115485	0.115485
9	1000	64	0.0010	0.170059	0.170059
10	1000	64	0.0100	0.179711	0.179711
11	1000	64	0.1000	0.372760	0.372760
12	1000	128	0.0001	0.140115	0.140115
13	1000	128	0.0010	0.182383	0.182383
14	1000	128	0.0100	0.287702	0.287702
15	1000	128	0.1000	0.342920	0.342920
16	2000	16	0.0001	0.115710	0.115710
17	2000	16	0.0010	0.132527	0.132527
18	2000	16	0.0100	0.142864	0.142864
19	2000	16	0.1000	0.473536	0.473536
20	2000	32	0.0001	0.115216	0.115216
21	2000	32	0.0010	0.137077	0.137077
22	2000	32	0.0100	0.233104	0.233104
23	2000	32	0.1000	0.332725	0.332725
24	2000	64	0.0001	0.137378	0.137378
25	2000	64	0.0010	0.151414	0.151414
26	2000	64	0.0100	0.216704	0.216704
27	2000	64	0.1000	0.346320	0.346320
28	2000	128	0.0001	0.131090	0.131090
29	2000	128	0.0010	0.191413	0.191413
30	2000	128	0.0100	0.248794	0.248794
31	2000	128	0.1000	0.345353	0.345353

Best Hyperparameters Found:
epochs 1000.000000
batch_size 32.000000
learning_rate 0.000100
loss 0.107589
mae 0.107589
Name: 4, dtype: float64

Training final model with best parameters...

Epoch 1/1000

C:\Users\barre\my_new_env2\Lib\site-packages\keras\src\layers\core\dense.py:92:
UserWarning: Do not pass an `input\_shape`/`input\_dim` argument to a layer. When
using Sequential models, prefer using an `Input(shape)` object as the first
layer in the model instead.

super().__init__(activity_regularizer=activity_regularizer, **kwargs)

57/57 1s 7ms/step - loss:

5.2940 - mae: 5.2940 - val_loss: 4.4075 - val_mae: 4.4075

Epoch 2/1000

57/57 0s 4ms/step - loss:

3.5318 - mae: 3.5318 - val_loss: 2.5151 - val_mae: 2.5151

Epoch 3/1000

57/57 0s 4ms/step - loss:
 1.9812 - mae: 1.9812 - val_loss: 1.6596 - val_mae: 1.6596
 Epoch 4/1000
 57/57 0s 4ms/step - loss:
 1.3978 - mae: 1.3978 - val_loss: 1.1797 - val_mae: 1.1797
 Epoch 5/1000
 57/57 0s 4ms/step - loss:
 1.0126 - mae: 1.0126 - val_loss: 0.8215 - val_mae: 0.8215
 Epoch 6/1000
 57/57 0s 4ms/step - loss:
 0.7420 - mae: 0.7420 - val_loss: 0.6262 - val_mae: 0.6262
 Epoch 7/1000
 57/57 0s 4ms/step - loss:
 0.5486 - mae: 0.5486 - val_loss: 0.4947 - val_mae: 0.4947
 Epoch 8/1000
 57/57 0s 4ms/step - loss:
 0.4152 - mae: 0.4152 - val_loss: 0.4036 - val_mae: 0.4036
 Epoch 9/1000
 57/57 0s 4ms/step - loss:
 0.3404 - mae: 0.3404 - val_loss: 0.3745 - val_mae: 0.3745
 Epoch 10/1000
 57/57 0s 4ms/step - loss:
 0.3065 - mae: 0.3065 - val_loss: 0.3175 - val_mae: 0.3175
 Epoch 11/1000
 57/57 0s 4ms/step - loss:
 0.2807 - mae: 0.2807 - val_loss: 0.3350 - val_mae: 0.3350
 Epoch 12/1000
 57/57 0s 4ms/step - loss:
 0.2641 - mae: 0.2641 - val_loss: 0.2934 - val_mae: 0.2934
 Epoch 13/1000
 57/57 0s 4ms/step - loss:
 0.2524 - mae: 0.2524 - val_loss: 0.2752 - val_mae: 0.2752
 Epoch 14/1000
 57/57 0s 4ms/step - loss:
 0.2390 - mae: 0.2390 - val_loss: 0.2440 - val_mae: 0.2440
 Epoch 15/1000
 57/57 0s 4ms/step - loss:
 0.2284 - mae: 0.2284 - val_loss: 0.2533 - val_mae: 0.2533
 Epoch 16/1000
 57/57 0s 4ms/step - loss:
 0.2209 - mae: 0.2209 - val_loss: 0.2340 - val_mae: 0.2340
 Epoch 17/1000
 57/57 0s 4ms/step - loss:
 0.2147 - mae: 0.2147 - val_loss: 0.2207 - val_mae: 0.2207
 Epoch 18/1000
 57/57 0s 4ms/step - loss:
 0.2065 - mae: 0.2065 - val_loss: 0.2327 - val_mae: 0.2327
 Epoch 19/1000

57/57 0s 4ms/step - loss:
 0.2017 - mae: 0.2017 - val_loss: 0.2103 - val_mae: 0.2103
 Epoch 20/1000
 57/57 0s 4ms/step - loss:
 0.1961 - mae: 0.1961 - val_loss: 0.2172 - val_mae: 0.2172
 Epoch 21/1000
 57/57 0s 4ms/step - loss:
 0.1954 - mae: 0.1954 - val_loss: 0.2238 - val_mae: 0.2238
 Epoch 22/1000
 57/57 0s 4ms/step - loss:
 0.1932 - mae: 0.1932 - val_loss: 0.2089 - val_mae: 0.2089
 Epoch 23/1000
 57/57 0s 4ms/step - loss:
 0.1918 - mae: 0.1918 - val_loss: 0.2081 - val_mae: 0.2081
 Epoch 24/1000
 57/57 0s 4ms/step - loss:
 0.1833 - mae: 0.1833 - val_loss: 0.2052 - val_mae: 0.2052
 Epoch 25/1000
 57/57 0s 4ms/step - loss:
 0.1818 - mae: 0.1818 - val_loss: 0.1929 - val_mae: 0.1929
 Epoch 26/1000
 57/57 0s 4ms/step - loss:
 0.1763 - mae: 0.1763 - val_loss: 0.1934 - val_mae: 0.1934
 Epoch 27/1000
 57/57 0s 4ms/step - loss:
 0.1783 - mae: 0.1783 - val_loss: 0.2605 - val_mae: 0.2605
 Epoch 28/1000
 57/57 0s 4ms/step - loss:
 0.1779 - mae: 0.1779 - val_loss: 0.1853 - val_mae: 0.1853
 Epoch 29/1000
 57/57 0s 4ms/step - loss:
 0.1728 - mae: 0.1728 - val_loss: 0.2362 - val_mae: 0.2362
 Epoch 30/1000
 57/57 0s 4ms/step - loss:
 0.1745 - mae: 0.1745 - val_loss: 0.1833 - val_mae: 0.1833
 Epoch 31/1000
 57/57 0s 4ms/step - loss:
 0.1687 - mae: 0.1687 - val_loss: 0.1866 - val_mae: 0.1866
 Epoch 32/1000
 57/57 0s 4ms/step - loss:
 0.1730 - mae: 0.1730 - val_loss: 0.1772 - val_mae: 0.1772
 Epoch 33/1000
 57/57 0s 4ms/step - loss:
 0.1671 - mae: 0.1671 - val_loss: 0.1987 - val_mae: 0.1987
 Epoch 34/1000
 57/57 0s 4ms/step - loss:
 0.1693 - mae: 0.1693 - val_loss: 0.1755 - val_mae: 0.1755
 Epoch 35/1000

57/57 0s 4ms/step - loss:
 0.1628 - mae: 0.1628 - val_loss: 0.2094 - val_mae: 0.2094
 Epoch 36/1000
 57/57 0s 4ms/step - loss:
 0.1636 - mae: 0.1636 - val_loss: 0.2522 - val_mae: 0.2522
 Epoch 37/1000
 57/57 0s 4ms/step - loss:
 0.1606 - mae: 0.1606 - val_loss: 0.1914 - val_mae: 0.1914
 Epoch 38/1000
 57/57 0s 4ms/step - loss:
 0.1554 - mae: 0.1554 - val_loss: 0.1993 - val_mae: 0.1993
 Epoch 39/1000
 57/57 0s 4ms/step - loss:
 0.1616 - mae: 0.1616 - val_loss: 0.1697 - val_mae: 0.1697
 Epoch 40/1000
 57/57 0s 4ms/step - loss:
 0.1581 - mae: 0.1581 - val_loss: 0.1826 - val_mae: 0.1826
 Epoch 41/1000
 57/57 0s 4ms/step - loss:
 0.1556 - mae: 0.1556 - val_loss: 0.1697 - val_mae: 0.1697
 Epoch 42/1000
 57/57 0s 4ms/step - loss:
 0.1568 - mae: 0.1568 - val_loss: 0.1917 - val_mae: 0.1917
 Epoch 43/1000
 57/57 0s 4ms/step - loss:
 0.1555 - mae: 0.1555 - val_loss: 0.1945 - val_mae: 0.1945
 Epoch 44/1000
 57/57 0s 4ms/step - loss:
 0.1566 - mae: 0.1566 - val_loss: 0.2155 - val_mae: 0.2155
 Epoch 45/1000
 57/57 0s 4ms/step - loss:
 0.1530 - mae: 0.1530 - val_loss: 0.2212 - val_mae: 0.2212
 Epoch 46/1000
 57/57 0s 4ms/step - loss:
 0.1579 - mae: 0.1579 - val_loss: 0.1661 - val_mae: 0.1661
 Epoch 47/1000
 57/57 0s 4ms/step - loss:
 0.1541 - mae: 0.1541 - val_loss: 0.1952 - val_mae: 0.1952
 Epoch 48/1000
 57/57 0s 4ms/step - loss:
 0.1501 - mae: 0.1501 - val_loss: 0.1854 - val_mae: 0.1854
 Epoch 49/1000
 57/57 0s 4ms/step - loss:
 0.1518 - mae: 0.1518 - val_loss: 0.1736 - val_mae: 0.1736
 Epoch 50/1000
 57/57 0s 4ms/step - loss:
 0.1501 - mae: 0.1501 - val_loss: 0.2182 - val_mae: 0.2182
 Epoch 51/1000


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57/57          0s 4ms/step - loss:
0.1504 - mae: 0.1504 - val_loss: 0.1925 - val_mae: 0.1925
Epoch 52/1000
57/57          0s 4ms/step - loss:
0.1514 - mae: 0.1514 - val_loss: 0.1733 - val_mae: 0.1733
Epoch 53/1000
57/57          0s 4ms/step - loss:
0.1481 - mae: 0.1481 - val_loss: 0.1706 - val_mae: 0.1706
Epoch 54/1000
57/57          0s 4ms/step - loss:
0.1486 - mae: 0.1486 - val_loss: 0.2050 - val_mae: 0.2050
Epoch 55/1000
57/57          0s 4ms/step - loss:
0.1454 - mae: 0.1454 - val_loss: 0.2005 - val_mae: 0.2005
Epoch 56/1000
57/57          0s 4ms/step - loss:
0.1454 - mae: 0.1454 - val_loss: 0.1985 - val_mae: 0.1985
Epoch 57/1000
57/57          0s 4ms/step - loss:
0.1441 - mae: 0.1441 - val_loss: 0.1867 - val_mae: 0.1867
Epoch 58/1000
57/57          0s 4ms/step - loss:
0.1479 - mae: 0.1479 - val_loss: 0.1651 - val_mae: 0.1651
Epoch 59/1000
57/57          0s 4ms/step - loss:
0.1422 - mae: 0.1422 - val_loss: 0.1558 - val_mae: 0.1558
Epoch 60/1000
57/57          0s 4ms/step - loss:
0.1389 - mae: 0.1389 - val_loss: 0.1821 - val_mae: 0.1821
Epoch 61/1000
57/57          0s 4ms/step - loss:
0.1400 - mae: 0.1400 - val_loss: 0.1633 - val_mae: 0.1633
Epoch 62/1000
57/57          0s 4ms/step - loss:
0.1451 - mae: 0.1451 - val_loss: 0.1610 - val_mae: 0.1610
Epoch 63/1000
57/57          0s 4ms/step - loss:
0.1419 - mae: 0.1419 - val_loss: 0.1713 - val_mae: 0.1713
Epoch 64/1000
57/57          0s 4ms/step - loss:
0.1405 - mae: 0.1405 - val_loss: 0.1581 - val_mae: 0.1581
Epoch 65/1000
57/57          0s 4ms/step - loss:
0.1422 - mae: 0.1422 - val_loss: 0.1652 - val_mae: 0.1652
Epoch 66/1000
57/57          0s 4ms/step - loss:
0.1407 - mae: 0.1407 - val_loss: 0.1931 - val_mae: 0.1931
Epoch 67/1000

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57/57 0s 4ms/step - loss:
 0.1419 - mae: 0.1419 - val_loss: 0.1661 - val_mae: 0.1661
 Epoch 68/1000
 57/57 0s 4ms/step - loss:
 0.1383 - mae: 0.1383 - val_loss: 0.1686 - val_mae: 0.1686
 Epoch 69/1000
 57/57 0s 4ms/step - loss:
 0.1408 - mae: 0.1408 - val_loss: 0.1549 - val_mae: 0.1549
 Epoch 70/1000
 57/57 0s 4ms/step - loss:
 0.1364 - mae: 0.1364 - val_loss: 0.1601 - val_mae: 0.1601
 Epoch 71/1000
 57/57 0s 4ms/step - loss:
 0.1392 - mae: 0.1392 - val_loss: 0.2158 - val_mae: 0.2158
 Epoch 72/1000
 57/57 0s 4ms/step - loss:
 0.1389 - mae: 0.1389 - val_loss: 0.2190 - val_mae: 0.2190
 Epoch 73/1000
 57/57 0s 4ms/step - loss:
 0.1354 - mae: 0.1354 - val_loss: 0.1626 - val_mae: 0.1626
 Epoch 74/1000
 57/57 0s 4ms/step - loss:
 0.1334 - mae: 0.1334 - val_loss: 0.1819 - val_mae: 0.1819
 Epoch 75/1000
 57/57 0s 4ms/step - loss:
 0.1367 - mae: 0.1367 - val_loss: 0.1752 - val_mae: 0.1752
 Epoch 76/1000
 57/57 0s 4ms/step - loss:
 0.1344 - mae: 0.1344 - val_loss: 0.1683 - val_mae: 0.1683
 Epoch 77/1000
 57/57 0s 4ms/step - loss:
 0.1353 - mae: 0.1353 - val_loss: 0.1506 - val_mae: 0.1506
 Epoch 78/1000
 57/57 0s 4ms/step - loss:
 0.1360 - mae: 0.1360 - val_loss: 0.1886 - val_mae: 0.1886
 Epoch 79/1000
 57/57 0s 4ms/step - loss:
 0.1298 - mae: 0.1298 - val_loss: 0.1821 - val_mae: 0.1821
 Epoch 80/1000
 57/57 0s 4ms/step - loss:
 0.1364 - mae: 0.1364 - val_loss: 0.1503 - val_mae: 0.1503
 Epoch 81/1000
 57/57 0s 4ms/step - loss:
 0.1311 - mae: 0.1311 - val_loss: 0.1458 - val_mae: 0.1458
 Epoch 82/1000
 57/57 0s 4ms/step - loss:
 0.1343 - mae: 0.1343 - val_loss: 0.1630 - val_mae: 0.1630
 Epoch 83/1000

57/57 0s 4ms/step - loss:
 0.1326 - mae: 0.1326 - val_loss: 0.1518 - val_mae: 0.1518
 Epoch 84/1000
 57/57 0s 4ms/step - loss:
 0.1324 - mae: 0.1324 - val_loss: 0.1668 - val_mae: 0.1668
 Epoch 85/1000
 57/57 0s 4ms/step - loss:
 0.1329 - mae: 0.1329 - val_loss: 0.1520 - val_mae: 0.1520
 Epoch 86/1000
 57/57 0s 4ms/step - loss:
 0.1278 - mae: 0.1278 - val_loss: 0.2175 - val_mae: 0.2175
 Epoch 87/1000
 57/57 0s 4ms/step - loss:
 0.1315 - mae: 0.1315 - val_loss: 0.1463 - val_mae: 0.1463
 Epoch 88/1000
 57/57 0s 4ms/step - loss:
 0.1311 - mae: 0.1311 - val_loss: 0.1691 - val_mae: 0.1691
 Epoch 89/1000
 57/57 0s 4ms/step - loss:
 0.1314 - mae: 0.1314 - val_loss: 0.1676 - val_mae: 0.1676
 Epoch 90/1000
 57/57 0s 4ms/step - loss:
 0.1309 - mae: 0.1309 - val_loss: 0.1449 - val_mae: 0.1449
 Epoch 91/1000
 57/57 0s 4ms/step - loss:
 0.1268 - mae: 0.1268 - val_loss: 0.1664 - val_mae: 0.1664
 Epoch 92/1000
 57/57 0s 4ms/step - loss:
 0.1283 - mae: 0.1283 - val_loss: 0.1858 - val_mae: 0.1858
 Epoch 93/1000
 57/57 0s 4ms/step - loss:
 0.1268 - mae: 0.1268 - val_loss: 0.1748 - val_mae: 0.1748
 Epoch 94/1000
 57/57 0s 4ms/step - loss:
 0.1286 - mae: 0.1286 - val_loss: 0.1450 - val_mae: 0.1450
 Epoch 95/1000
 57/57 0s 4ms/step - loss:
 0.1273 - mae: 0.1273 - val_loss: 0.1748 - val_mae: 0.1748
 Epoch 96/1000
 57/57 0s 4ms/step - loss:
 0.1296 - mae: 0.1296 - val_loss: 0.1774 - val_mae: 0.1774
 Epoch 97/1000
 57/57 0s 4ms/step - loss:
 0.1274 - mae: 0.1274 - val_loss: 0.1630 - val_mae: 0.1630
 Epoch 98/1000
 57/57 0s 4ms/step - loss:
 0.1277 - mae: 0.1277 - val_loss: 0.1888 - val_mae: 0.1888
 Epoch 99/1000

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57/57          0s 4ms/step - loss:
0.1242 - mae: 0.1242 - val_loss: 0.1619 - val_mae: 0.1619
Epoch 100/1000
57/57          0s 4ms/step - loss:
0.1301 - mae: 0.1301 - val_loss: 0.1456 - val_mae: 0.1456
Epoch 101/1000
57/57          0s 4ms/step - loss:
0.1264 - mae: 0.1264 - val_loss: 0.1404 - val_mae: 0.1404
Epoch 102/1000
57/57          0s 4ms/step - loss:
0.1263 - mae: 0.1263 - val_loss: 0.1587 - val_mae: 0.1587
Epoch 103/1000
57/57          0s 4ms/step - loss:
0.1251 - mae: 0.1251 - val_loss: 0.1532 - val_mae: 0.1532
Epoch 104/1000
57/57          0s 5ms/step - loss:
0.1272 - mae: 0.1272 - val_loss: 0.1744 - val_mae: 0.1744
Epoch 105/1000
57/57          0s 4ms/step - loss:
0.1226 - mae: 0.1226 - val_loss: 0.1425 - val_mae: 0.1425
Epoch 106/1000
57/57          0s 4ms/step - loss:
0.1273 - mae: 0.1273 - val_loss: 0.1735 - val_mae: 0.1735
Epoch 107/1000
57/57          0s 4ms/step - loss:
0.1215 - mae: 0.1215 - val_loss: 0.1439 - val_mae: 0.1439
Epoch 108/1000
57/57          0s 4ms/step - loss:
0.1251 - mae: 0.1251 - val_loss: 0.1571 - val_mae: 0.1571
Epoch 109/1000
57/57          0s 4ms/step - loss:
0.1204 - mae: 0.1204 - val_loss: 0.1776 - val_mae: 0.1776
Epoch 110/1000
57/57          0s 4ms/step - loss:
0.1222 - mae: 0.1222 - val_loss: 0.1440 - val_mae: 0.1440
Epoch 111/1000
57/57          0s 4ms/step - loss:
0.1222 - mae: 0.1222 - val_loss: 0.1492 - val_mae: 0.1492
Epoch 112/1000
57/57          0s 4ms/step - loss:
0.1252 - mae: 0.1252 - val_loss: 0.1848 - val_mae: 0.1848
Epoch 113/1000
57/57          0s 4ms/step - loss:
0.1203 - mae: 0.1203 - val_loss: 0.2111 - val_mae: 0.2111
Epoch 114/1000
57/57          0s 4ms/step - loss:
0.1266 - mae: 0.1266 - val_loss: 0.1697 - val_mae: 0.1697
Epoch 115/1000

```

```

57/57          0s 4ms/step - loss:
0.1260 - mae: 0.1260 - val_loss: 0.1705 - val_mae: 0.1705
Epoch 116/1000
57/57          0s 4ms/step - loss:
0.1181 - mae: 0.1181 - val_loss: 0.1484 - val_mae: 0.1484
Epoch 117/1000
57/57          0s 4ms/step - loss:
0.1244 - mae: 0.1244 - val_loss: 0.1538 - val_mae: 0.1538
Epoch 118/1000
57/57          0s 4ms/step - loss:
0.1215 - mae: 0.1215 - val_loss: 0.1560 - val_mae: 0.1560
Epoch 119/1000
57/57          0s 4ms/step - loss:
0.1191 - mae: 0.1191 - val_loss: 0.1505 - val_mae: 0.1505
Epoch 120/1000
57/57          0s 4ms/step - loss:
0.1152 - mae: 0.1152 - val_loss: 0.1454 - val_mae: 0.1454
Epoch 121/1000
57/57          0s 4ms/step - loss:
0.1187 - mae: 0.1187 - val_loss: 0.1908 - val_mae: 0.1908
Epoch 122/1000
57/57          0s 4ms/step - loss:
0.1209 - mae: 0.1209 - val_loss: 0.1656 - val_mae: 0.1656
Epoch 123/1000
57/57          0s 4ms/step - loss:
0.1203 - mae: 0.1203 - val_loss: 0.1391 - val_mae: 0.1391
Epoch 124/1000
57/57          0s 4ms/step - loss:
0.1153 - mae: 0.1153 - val_loss: 0.1476 - val_mae: 0.1476
Epoch 125/1000
57/57          0s 4ms/step - loss:
0.1202 - mae: 0.1202 - val_loss: 0.1599 - val_mae: 0.1599
Epoch 126/1000
57/57          0s 4ms/step - loss:
0.1206 - mae: 0.1206 - val_loss: 0.1359 - val_mae: 0.1359
Epoch 127/1000
57/57          0s 4ms/step - loss:
0.1167 - mae: 0.1167 - val_loss: 0.1567 - val_mae: 0.1567
Epoch 128/1000
57/57          0s 4ms/step - loss:
0.1200 - mae: 0.1200 - val_loss: 0.1337 - val_mae: 0.1337
Epoch 129/1000
57/57          0s 4ms/step - loss:
0.1170 - mae: 0.1170 - val_loss: 0.1350 - val_mae: 0.1350
Epoch 130/1000
57/57          0s 4ms/step - loss:
0.1179 - mae: 0.1179 - val_loss: 0.1476 - val_mae: 0.1476
Epoch 131/1000

```

57/57 0s 4ms/step - loss:
 0.1164 - mae: 0.1164 - val_loss: 0.1327 - val_mae: 0.1327
 Epoch 132/1000
 57/57 0s 4ms/step - loss:
 0.1191 - mae: 0.1191 - val_loss: 0.1321 - val_mae: 0.1321
 Epoch 133/1000
 57/57 0s 4ms/step - loss:
 0.1152 - mae: 0.1152 - val_loss: 0.1343 - val_mae: 0.1343
 Epoch 134/1000
 57/57 0s 4ms/step - loss:
 0.1167 - mae: 0.1167 - val_loss: 0.1490 - val_mae: 0.1490
 Epoch 135/1000
 57/57 0s 4ms/step - loss:
 0.1183 - mae: 0.1183 - val_loss: 0.1450 - val_mae: 0.1450
 Epoch 136/1000
 57/57 0s 4ms/step - loss:
 0.1164 - mae: 0.1164 - val_loss: 0.1760 - val_mae: 0.1760
 Epoch 137/1000
 57/57 0s 4ms/step - loss:
 0.1205 - mae: 0.1205 - val_loss: 0.1608 - val_mae: 0.1608
 Epoch 138/1000
 57/57 0s 4ms/step - loss:
 0.1167 - mae: 0.1167 - val_loss: 0.1872 - val_mae: 0.1872
 Epoch 139/1000
 57/57 0s 4ms/step - loss:
 0.1158 - mae: 0.1158 - val_loss: 0.1495 - val_mae: 0.1495
 Epoch 140/1000
 57/57 0s 4ms/step - loss:
 0.1165 - mae: 0.1165 - val_loss: 0.1512 - val_mae: 0.1512
 Epoch 141/1000
 57/57 0s 4ms/step - loss:
 0.1149 - mae: 0.1149 - val_loss: 0.1579 - val_mae: 0.1579
 Epoch 142/1000
 57/57 0s 4ms/step - loss:
 0.1155 - mae: 0.1155 - val_loss: 0.1622 - val_mae: 0.1622
 Epoch 143/1000
 57/57 0s 4ms/step - loss:
 0.1144 - mae: 0.1144 - val_loss: 0.1316 - val_mae: 0.1316
 Epoch 144/1000
 57/57 0s 4ms/step - loss:
 0.1145 - mae: 0.1145 - val_loss: 0.1368 - val_mae: 0.1368
 Epoch 145/1000
 57/57 0s 4ms/step - loss:
 0.1166 - mae: 0.1166 - val_loss: 0.1463 - val_mae: 0.1463
 Epoch 146/1000
 57/57 0s 4ms/step - loss:
 0.1148 - mae: 0.1148 - val_loss: 0.1443 - val_mae: 0.1443
 Epoch 147/1000

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57/57          0s 4ms/step - loss:
0.1147 - mae: 0.1147 - val_loss: 0.1637 - val_mae: 0.1637
Epoch 148/1000
57/57          0s 4ms/step - loss:
0.1141 - mae: 0.1141 - val_loss: 0.1645 - val_mae: 0.1645
Epoch 149/1000
57/57          0s 4ms/step - loss:
0.1173 - mae: 0.1173 - val_loss: 0.1489 - val_mae: 0.1489
Epoch 150/1000
57/57          0s 5ms/step - loss:
0.1118 - mae: 0.1118 - val_loss: 0.1653 - val_mae: 0.1653
Epoch 151/1000
57/57          0s 4ms/step - loss:
0.1147 - mae: 0.1147 - val_loss: 0.1591 - val_mae: 0.1591
Epoch 152/1000
57/57          0s 4ms/step - loss:
0.1165 - mae: 0.1165 - val_loss: 0.1454 - val_mae: 0.1454
Epoch 153/1000
57/57          0s 4ms/step - loss:
0.1120 - mae: 0.1120 - val_loss: 0.1408 - val_mae: 0.1408
Epoch 154/1000
57/57          0s 4ms/step - loss:
0.1123 - mae: 0.1123 - val_loss: 0.1565 - val_mae: 0.1565
Epoch 155/1000
57/57          0s 4ms/step - loss:
0.1140 - mae: 0.1140 - val_loss: 0.1700 - val_mae: 0.1700
Epoch 156/1000
57/57          0s 4ms/step - loss:
0.1164 - mae: 0.1164 - val_loss: 0.1562 - val_mae: 0.1562
Epoch 157/1000
57/57          0s 4ms/step - loss:
0.1104 - mae: 0.1104 - val_loss: 0.1409 - val_mae: 0.1409
Epoch 158/1000
57/57          0s 4ms/step - loss:
0.1128 - mae: 0.1128 - val_loss: 0.1424 - val_mae: 0.1424
Epoch 159/1000
57/57          0s 4ms/step - loss:
0.1144 - mae: 0.1144 - val_loss: 0.1297 - val_mae: 0.1297
Epoch 160/1000
57/57          0s 4ms/step - loss:
0.1127 - mae: 0.1127 - val_loss: 0.1766 - val_mae: 0.1766
Epoch 161/1000
57/57          0s 4ms/step - loss:
0.1151 - mae: 0.1151 - val_loss: 0.1859 - val_mae: 0.1859
Epoch 162/1000
57/57          0s 4ms/step - loss:
0.1103 - mae: 0.1103 - val_loss: 0.1645 - val_mae: 0.1645
Epoch 163/1000

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57/57          0s 4ms/step - loss:
0.1142 - mae: 0.1142 - val_loss: 0.1542 - val_mae: 0.1542
Epoch 164/1000
57/57          0s 4ms/step - loss:
0.1121 - mae: 0.1121 - val_loss: 0.1812 - val_mae: 0.1812
Epoch 165/1000
57/57          0s 4ms/step - loss:
0.1112 - mae: 0.1112 - val_loss: 0.1461 - val_mae: 0.1461
Epoch 166/1000
57/57          0s 4ms/step - loss:
0.1101 - mae: 0.1101 - val_loss: 0.1487 - val_mae: 0.1487
Epoch 167/1000
57/57          0s 4ms/step - loss:
0.1109 - mae: 0.1109 - val_loss: 0.1340 - val_mae: 0.1340
Epoch 168/1000
57/57          0s 4ms/step - loss:
0.1107 - mae: 0.1107 - val_loss: 0.1455 - val_mae: 0.1455
Epoch 169/1000
57/57          0s 4ms/step - loss:
0.1120 - mae: 0.1120 - val_loss: 0.1291 - val_mae: 0.1291
Epoch 170/1000
57/57          0s 4ms/step - loss:
0.1135 - mae: 0.1135 - val_loss: 0.1363 - val_mae: 0.1363
Epoch 171/1000
57/57          0s 4ms/step - loss:
0.1105 - mae: 0.1105 - val_loss: 0.1248 - val_mae: 0.1248
Epoch 172/1000
57/57          0s 4ms/step - loss:
0.1096 - mae: 0.1096 - val_loss: 0.1268 - val_mae: 0.1268
Epoch 173/1000
57/57          0s 4ms/step - loss:
0.1069 - mae: 0.1069 - val_loss: 0.1443 - val_mae: 0.1443
Epoch 174/1000
57/57          0s 4ms/step - loss:
0.1123 - mae: 0.1123 - val_loss: 0.1712 - val_mae: 0.1712
Epoch 175/1000
57/57          0s 4ms/step - loss:
0.1116 - mae: 0.1116 - val_loss: 0.1277 - val_mae: 0.1277
Epoch 176/1000
57/57          0s 4ms/step - loss:
0.1093 - mae: 0.1093 - val_loss: 0.1690 - val_mae: 0.1690
Epoch 177/1000
57/57          0s 4ms/step - loss:
0.1121 - mae: 0.1121 - val_loss: 0.1268 - val_mae: 0.1268
Epoch 178/1000
57/57          0s 4ms/step - loss:
0.1144 - mae: 0.1144 - val_loss: 0.1286 - val_mae: 0.1286
Epoch 179/1000

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57/57          0s 4ms/step - loss:
0.1099 - mae: 0.1099 - val_loss: 0.1706 - val_mae: 0.1706
Epoch 180/1000
57/57          0s 4ms/step - loss:
0.1113 - mae: 0.1113 - val_loss: 0.1360 - val_mae: 0.1360
Epoch 181/1000
57/57          0s 4ms/step - loss:
0.1063 - mae: 0.1063 - val_loss: 0.1420 - val_mae: 0.1420
Epoch 182/1000
57/57          0s 4ms/step - loss:
0.1089 - mae: 0.1089 - val_loss: 0.1422 - val_mae: 0.1422
Epoch 183/1000
57/57          0s 4ms/step - loss:
0.1065 - mae: 0.1065 - val_loss: 0.1362 - val_mae: 0.1362
Epoch 184/1000
57/57          0s 4ms/step - loss:
0.1084 - mae: 0.1084 - val_loss: 0.1226 - val_mae: 0.1226
Epoch 185/1000
57/57          0s 4ms/step - loss:
0.1081 - mae: 0.1081 - val_loss: 0.1344 - val_mae: 0.1344
Epoch 186/1000
57/57          0s 4ms/step - loss:
0.1097 - mae: 0.1097 - val_loss: 0.1732 - val_mae: 0.1732
Epoch 187/1000
57/57          0s 4ms/step - loss:
0.1083 - mae: 0.1083 - val_loss: 0.1406 - val_mae: 0.1406
Epoch 188/1000
57/57          0s 4ms/step - loss:
0.1069 - mae: 0.1069 - val_loss: 0.1285 - val_mae: 0.1285
Epoch 189/1000
57/57          0s 4ms/step - loss:
0.1061 - mae: 0.1061 - val_loss: 0.1331 - val_mae: 0.1331
Epoch 190/1000
57/57          0s 4ms/step - loss:
0.1074 - mae: 0.1074 - val_loss: 0.1306 - val_mae: 0.1306
Epoch 191/1000
57/57          0s 4ms/step - loss:
0.1085 - mae: 0.1085 - val_loss: 0.1303 - val_mae: 0.1303
Epoch 192/1000
57/57          0s 4ms/step - loss:
0.1068 - mae: 0.1068 - val_loss: 0.1329 - val_mae: 0.1329
Epoch 193/1000
57/57          0s 4ms/step - loss:
0.1062 - mae: 0.1062 - val_loss: 0.1430 - val_mae: 0.1430
Epoch 194/1000
57/57          0s 4ms/step - loss:
0.1061 - mae: 0.1061 - val_loss: 0.1671 - val_mae: 0.1671
Epoch 195/1000

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57/57          0s 4ms/step - loss:
0.1074 - mae: 0.1074 - val_loss: 0.1386 - val_mae: 0.1386
Epoch 196/1000
57/57          0s 4ms/step - loss:
0.1076 - mae: 0.1076 - val_loss: 0.1576 - val_mae: 0.1576
Epoch 197/1000
57/57          0s 4ms/step - loss:
0.1080 - mae: 0.1080 - val_loss: 0.1339 - val_mae: 0.1339
Epoch 198/1000
57/57          0s 4ms/step - loss:
0.1053 - mae: 0.1053 - val_loss: 0.1431 - val_mae: 0.1431
Epoch 199/1000
57/57          0s 4ms/step - loss:
0.1089 - mae: 0.1089 - val_loss: 0.1429 - val_mae: 0.1429
Epoch 200/1000
57/57          0s 4ms/step - loss:
0.1071 - mae: 0.1071 - val_loss: 0.1533 - val_mae: 0.1533
Epoch 201/1000
57/57          0s 4ms/step - loss:
0.1031 - mae: 0.1031 - val_loss: 0.1216 - val_mae: 0.1216
Epoch 202/1000
57/57          0s 4ms/step - loss:
0.1072 - mae: 0.1072 - val_loss: 0.1602 - val_mae: 0.1602
Epoch 203/1000
57/57          0s 4ms/step - loss:
0.1032 - mae: 0.1032 - val_loss: 0.1666 - val_mae: 0.1666
Epoch 204/1000
57/57          0s 4ms/step - loss:
0.1064 - mae: 0.1064 - val_loss: 0.1693 - val_mae: 0.1693
Epoch 205/1000
57/57          0s 4ms/step - loss:
0.1074 - mae: 0.1074 - val_loss: 0.1276 - val_mae: 0.1276
Epoch 206/1000
57/57          0s 4ms/step - loss:
0.1056 - mae: 0.1056 - val_loss: 0.1728 - val_mae: 0.1728
Epoch 207/1000
57/57          0s 4ms/step - loss:
0.1075 - mae: 0.1075 - val_loss: 0.1443 - val_mae: 0.1443
Epoch 208/1000
57/57          0s 4ms/step - loss:
0.1031 - mae: 0.1031 - val_loss: 0.1405 - val_mae: 0.1405
Epoch 209/1000
57/57          0s 4ms/step - loss:
0.1072 - mae: 0.1072 - val_loss: 0.1782 - val_mae: 0.1782
Epoch 210/1000
57/57          0s 4ms/step - loss:
0.1063 - mae: 0.1063 - val_loss: 0.1186 - val_mae: 0.1186
Epoch 211/1000

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57/57          0s 4ms/step - loss:
0.1032 - mae: 0.1032 - val_loss: 0.1514 - val_mae: 0.1514
Epoch 212/1000
57/57          0s 4ms/step - loss:
0.1040 - mae: 0.1040 - val_loss: 0.1295 - val_mae: 0.1295
Epoch 213/1000
57/57          0s 4ms/step - loss:
0.1072 - mae: 0.1072 - val_loss: 0.1538 - val_mae: 0.1538
Epoch 214/1000
57/57          0s 4ms/step - loss:
0.1083 - mae: 0.1083 - val_loss: 0.1496 - val_mae: 0.1496
Epoch 215/1000
57/57          0s 4ms/step - loss:
0.1008 - mae: 0.1008 - val_loss: 0.1499 - val_mae: 0.1499
Epoch 216/1000
57/57          0s 4ms/step - loss:
0.1055 - mae: 0.1055 - val_loss: 0.1339 - val_mae: 0.1339
Epoch 217/1000
57/57          0s 4ms/step - loss:
0.1049 - mae: 0.1049 - val_loss: 0.1595 - val_mae: 0.1595
Epoch 218/1000
57/57          0s 4ms/step - loss:
0.1040 - mae: 0.1040 - val_loss: 0.1384 - val_mae: 0.1384
Epoch 219/1000
57/57          0s 4ms/step - loss:
0.1035 - mae: 0.1035 - val_loss: 0.1599 - val_mae: 0.1599
Epoch 220/1000
57/57          0s 4ms/step - loss:
0.1043 - mae: 0.1043 - val_loss: 0.1510 - val_mae: 0.1510
Epoch 221/1000
57/57          0s 4ms/step - loss:
0.1081 - mae: 0.1081 - val_loss: 0.1511 - val_mae: 0.1511
Epoch 222/1000
57/57          0s 4ms/step - loss:
0.1021 - mae: 0.1021 - val_loss: 0.1212 - val_mae: 0.1212
Epoch 223/1000
57/57          0s 4ms/step - loss:
0.1035 - mae: 0.1035 - val_loss: 0.1261 - val_mae: 0.1261
Epoch 224/1000
57/57          0s 4ms/step - loss:
0.1028 - mae: 0.1028 - val_loss: 0.1423 - val_mae: 0.1423
Epoch 225/1000
57/57          0s 4ms/step - loss:
0.1070 - mae: 0.1070 - val_loss: 0.1430 - val_mae: 0.1430
Epoch 226/1000
57/57          0s 4ms/step - loss:
0.1004 - mae: 0.1004 - val_loss: 0.1695 - val_mae: 0.1695
Epoch 227/1000

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57/57          0s 4ms/step - loss:
0.1066 - mae: 0.1066 - val_loss: 0.1408 - val_mae: 0.1408
Epoch 228/1000
57/57          0s 4ms/step - loss:
0.1038 - mae: 0.1038 - val_loss: 0.1669 - val_mae: 0.1669
Epoch 229/1000
57/57          0s 4ms/step - loss:
0.1011 - mae: 0.1011 - val_loss: 0.1787 - val_mae: 0.1787
Epoch 230/1000
57/57          0s 4ms/step - loss:
0.1003 - mae: 0.1003 - val_loss: 0.1628 - val_mae: 0.1628
Epoch 231/1000
57/57          0s 4ms/step - loss:
0.1027 - mae: 0.1027 - val_loss: 0.1166 - val_mae: 0.1166
Epoch 232/1000
57/57          0s 4ms/step - loss:
0.1046 - mae: 0.1046 - val_loss: 0.1207 - val_mae: 0.1207
Epoch 233/1000
57/57          0s 4ms/step - loss:
0.1026 - mae: 0.1026 - val_loss: 0.1263 - val_mae: 0.1263
Epoch 234/1000
57/57          0s 4ms/step - loss:
0.1032 - mae: 0.1032 - val_loss: 0.1554 - val_mae: 0.1554
Epoch 235/1000
57/57          0s 4ms/step - loss:
0.1003 - mae: 0.1003 - val_loss: 0.1765 - val_mae: 0.1765
Epoch 236/1000
57/57          0s 4ms/step - loss:
0.1037 - mae: 0.1037 - val_loss: 0.1211 - val_mae: 0.1211
Epoch 237/1000
57/57          0s 4ms/step - loss:
0.1019 - mae: 0.1019 - val_loss: 0.1235 - val_mae: 0.1235
Epoch 238/1000
57/57          0s 4ms/step - loss:
0.1018 - mae: 0.1018 - val_loss: 0.1336 - val_mae: 0.1336
Epoch 239/1000
57/57          0s 4ms/step - loss:
0.1008 - mae: 0.1008 - val_loss: 0.1749 - val_mae: 0.1749
Epoch 240/1000
57/57          0s 4ms/step - loss:
0.1000 - mae: 0.1000 - val_loss: 0.1313 - val_mae: 0.1313
Epoch 241/1000
57/57          0s 4ms/step - loss:
0.1017 - mae: 0.1017 - val_loss: 0.1219 - val_mae: 0.1219
Epoch 242/1000
57/57          0s 4ms/step - loss:
0.1011 - mae: 0.1011 - val_loss: 0.1388 - val_mae: 0.1388
Epoch 243/1000

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57/57 0s 4ms/step - loss:
 0.0987 - mae: 0.0987 - val_loss: 0.1338 - val_mae: 0.1338
 Epoch 244/1000
 57/57 0s 4ms/step - loss:
 0.1012 - mae: 0.1012 - val_loss: 0.1420 - val_mae: 0.1420
 Epoch 245/1000
 57/57 0s 4ms/step - loss:
 0.1012 - mae: 0.1012 - val_loss: 0.1320 - val_mae: 0.1320
 Epoch 246/1000
 57/57 0s 4ms/step - loss:
 0.1007 - mae: 0.1007 - val_loss: 0.1206 - val_mae: 0.1206
 Epoch 247/1000
 57/57 0s 4ms/step - loss:
 0.1008 - mae: 0.1008 - val_loss: 0.1388 - val_mae: 0.1388
 Epoch 248/1000
 57/57 0s 4ms/step - loss:
 0.0961 - mae: 0.0961 - val_loss: 0.1289 - val_mae: 0.1289
 Epoch 249/1000
 57/57 0s 4ms/step - loss:
 0.1031 - mae: 0.1031 - val_loss: 0.1199 - val_mae: 0.1199
 Epoch 250/1000
 57/57 0s 4ms/step - loss:
 0.0996 - mae: 0.0996 - val_loss: 0.1159 - val_mae: 0.1159
 Epoch 251/1000
 57/57 0s 4ms/step - loss:
 0.1008 - mae: 0.1008 - val_loss: 0.1228 - val_mae: 0.1228
 Epoch 252/1000
 57/57 0s 4ms/step - loss:
 0.0974 - mae: 0.0974 - val_loss: 0.1564 - val_mae: 0.1564
 Epoch 253/1000
 57/57 0s 4ms/step - loss:
 0.0990 - mae: 0.0990 - val_loss: 0.1180 - val_mae: 0.1180
 Epoch 254/1000
 57/57 0s 4ms/step - loss:
 0.1016 - mae: 0.1016 - val_loss: 0.1157 - val_mae: 0.1157
 Epoch 255/1000
 57/57 0s 4ms/step - loss:
 0.1015 - mae: 0.1015 - val_loss: 0.1258 - val_mae: 0.1258
 Epoch 256/1000
 57/57 0s 4ms/step - loss:
 0.1031 - mae: 0.1031 - val_loss: 0.1184 - val_mae: 0.1184
 Epoch 257/1000
 57/57 0s 4ms/step - loss:
 0.0976 - mae: 0.0976 - val_loss: 0.1411 - val_mae: 0.1411
 Epoch 258/1000
 57/57 0s 4ms/step - loss:
 0.1010 - mae: 0.1010 - val_loss: 0.1321 - val_mae: 0.1321
 Epoch 259/1000

57/57 0s 4ms/step - loss:
 0.1022 - mae: 0.1022 - val_loss: 0.1234 - val_mae: 0.1234
 Epoch 260/1000
 57/57 0s 4ms/step - loss:
 0.1000 - mae: 0.1000 - val_loss: 0.1205 - val_mae: 0.1205
 Epoch 261/1000
 57/57 0s 4ms/step - loss:
 0.1011 - mae: 0.1011 - val_loss: 0.1480 - val_mae: 0.1480
 Epoch 262/1000
 57/57 0s 4ms/step - loss:
 0.0967 - mae: 0.0967 - val_loss: 0.1311 - val_mae: 0.1311
 Epoch 263/1000
 57/57 0s 4ms/step - loss:
 0.1022 - mae: 0.1022 - val_loss: 0.1528 - val_mae: 0.1528
 Epoch 264/1000
 57/57 0s 4ms/step - loss:
 0.0973 - mae: 0.0973 - val_loss: 0.1635 - val_mae: 0.1635
 Epoch 265/1000
 57/57 0s 4ms/step - loss:
 0.1016 - mae: 0.1016 - val_loss: 0.1159 - val_mae: 0.1159
 Epoch 266/1000
 57/57 0s 4ms/step - loss:
 0.1006 - mae: 0.1006 - val_loss: 0.1273 - val_mae: 0.1273
 Epoch 267/1000
 57/57 0s 4ms/step - loss:
 0.0985 - mae: 0.0985 - val_loss: 0.1167 - val_mae: 0.1167
 Epoch 268/1000
 57/57 0s 4ms/step - loss:
 0.1004 - mae: 0.1004 - val_loss: 0.1292 - val_mae: 0.1292
 Epoch 269/1000
 57/57 0s 4ms/step - loss:
 0.0971 - mae: 0.0971 - val_loss: 0.1223 - val_mae: 0.1223
 Epoch 270/1000
 57/57 0s 4ms/step - loss:
 0.0986 - mae: 0.0986 - val_loss: 0.1162 - val_mae: 0.1162
 Epoch 271/1000
 57/57 0s 4ms/step - loss:
 0.0976 - mae: 0.0976 - val_loss: 0.1331 - val_mae: 0.1331
 Epoch 272/1000
 57/57 0s 4ms/step - loss:
 0.0991 - mae: 0.0991 - val_loss: 0.1290 - val_mae: 0.1290
 Epoch 273/1000
 57/57 0s 4ms/step - loss:
 0.0963 - mae: 0.0963 - val_loss: 0.1143 - val_mae: 0.1143
 Epoch 274/1000
 57/57 0s 4ms/step - loss:
 0.0989 - mae: 0.0989 - val_loss: 0.1335 - val_mae: 0.1335
 Epoch 275/1000

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57/57          0s 4ms/step - loss:
0.0975 - mae: 0.0975 - val_loss: 0.1114 - val_mae: 0.1114
Epoch 276/1000
57/57          0s 4ms/step - loss:
0.0965 - mae: 0.0965 - val_loss: 0.1790 - val_mae: 0.1790
Epoch 277/1000
57/57          0s 4ms/step - loss:
0.0993 - mae: 0.0993 - val_loss: 0.1119 - val_mae: 0.1119
Epoch 278/1000
57/57          0s 4ms/step - loss:
0.0961 - mae: 0.0961 - val_loss: 0.1247 - val_mae: 0.1247
Epoch 279/1000
57/57          0s 4ms/step - loss:
0.1005 - mae: 0.1005 - val_loss: 0.1279 - val_mae: 0.1279
Epoch 280/1000
57/57          0s 4ms/step - loss:
0.0976 - mae: 0.0976 - val_loss: 0.1389 - val_mae: 0.1389
Epoch 281/1000
57/57          0s 4ms/step - loss:
0.0988 - mae: 0.0988 - val_loss: 0.1378 - val_mae: 0.1378
Epoch 282/1000
57/57          0s 4ms/step - loss:
0.0979 - mae: 0.0979 - val_loss: 0.1282 - val_mae: 0.1282
Epoch 283/1000
57/57          0s 4ms/step - loss:
0.0955 - mae: 0.0955 - val_loss: 0.1272 - val_mae: 0.1272
Epoch 284/1000
57/57          0s 4ms/step - loss:
0.0997 - mae: 0.0997 - val_loss: 0.1463 - val_mae: 0.1463
Epoch 285/1000
57/57          0s 4ms/step - loss:
0.0965 - mae: 0.0965 - val_loss: 0.1442 - val_mae: 0.1442
Epoch 286/1000
57/57          0s 4ms/step - loss:
0.0940 - mae: 0.0940 - val_loss: 0.1577 - val_mae: 0.1577
Epoch 287/1000
57/57          0s 4ms/step - loss:
0.1001 - mae: 0.1001 - val_loss: 0.1272 - val_mae: 0.1272
Epoch 288/1000
57/57          0s 4ms/step - loss:
0.0971 - mae: 0.0971 - val_loss: 0.1130 - val_mae: 0.1130
Epoch 289/1000
57/57          0s 4ms/step - loss:
0.0981 - mae: 0.0981 - val_loss: 0.1122 - val_mae: 0.1122
Epoch 290/1000
57/57          0s 4ms/step - loss:
0.0915 - mae: 0.0915 - val_loss: 0.1090 - val_mae: 0.1090
Epoch 291/1000

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57/57 0s 4ms/step - loss:
 0.0977 - mae: 0.0977 - val_loss: 0.1364 - val_mae: 0.1364
 Epoch 292/1000
 57/57 0s 4ms/step - loss:
 0.0954 - mae: 0.0954 - val_loss: 0.1369 - val_mae: 0.1369
 Epoch 293/1000
 57/57 0s 4ms/step - loss:
 0.0976 - mae: 0.0976 - val_loss: 0.1123 - val_mae: 0.1123
 Epoch 294/1000
 57/57 0s 4ms/step - loss:
 0.0982 - mae: 0.0982 - val_loss: 0.1498 - val_mae: 0.1498
 Epoch 295/1000
 57/57 0s 4ms/step - loss:
 0.0970 - mae: 0.0970 - val_loss: 0.1184 - val_mae: 0.1184
 Epoch 296/1000
 57/57 0s 4ms/step - loss:
 0.0962 - mae: 0.0962 - val_loss: 0.1167 - val_mae: 0.1167
 Epoch 297/1000
 57/57 0s 4ms/step - loss:
 0.0972 - mae: 0.0972 - val_loss: 0.1471 - val_mae: 0.1471
 Epoch 298/1000
 57/57 0s 4ms/step - loss:
 0.0978 - mae: 0.0978 - val_loss: 0.1220 - val_mae: 0.1220
 Epoch 299/1000
 57/57 0s 4ms/step - loss:
 0.0959 - mae: 0.0959 - val_loss: 0.1307 - val_mae: 0.1307
 Epoch 300/1000
 57/57 0s 4ms/step - loss:
 0.0955 - mae: 0.0955 - val_loss: 0.1414 - val_mae: 0.1414
 Epoch 301/1000
 57/57 0s 4ms/step - loss:
 0.0944 - mae: 0.0944 - val_loss: 0.1266 - val_mae: 0.1266
 Epoch 302/1000
 57/57 0s 4ms/step - loss:
 0.0949 - mae: 0.0949 - val_loss: 0.1327 - val_mae: 0.1327
 Epoch 303/1000
 57/57 0s 4ms/step - loss:
 0.0944 - mae: 0.0944 - val_loss: 0.1461 - val_mae: 0.1461
 Epoch 304/1000
 57/57 0s 4ms/step - loss:
 0.0959 - mae: 0.0959 - val_loss: 0.1251 - val_mae: 0.1251
 Epoch 305/1000
 57/57 0s 4ms/step - loss:
 0.0959 - mae: 0.0959 - val_loss: 0.1315 - val_mae: 0.1315
 Epoch 306/1000
 57/57 0s 4ms/step - loss:
 0.0967 - mae: 0.0967 - val_loss: 0.1286 - val_mae: 0.1286
 Epoch 307/1000


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57/57          0s 4ms/step - loss:
0.0950 - mae: 0.0950 - val_loss: 0.1195 - val_mae: 0.1195
Epoch 308/1000
57/57          0s 4ms/step - loss:
0.0988 - mae: 0.0988 - val_loss: 0.1415 - val_mae: 0.1415
Epoch 309/1000
57/57          0s 4ms/step - loss:
0.0966 - mae: 0.0966 - val_loss: 0.1555 - val_mae: 0.1555
Epoch 310/1000
57/57          0s 4ms/step - loss:
0.0942 - mae: 0.0942 - val_loss: 0.1420 - val_mae: 0.1420
Epoch 311/1000
57/57          0s 4ms/step - loss:
0.0953 - mae: 0.0953 - val_loss: 0.1192 - val_mae: 0.1192
Epoch 312/1000
57/57          0s 4ms/step - loss:
0.0947 - mae: 0.0947 - val_loss: 0.1520 - val_mae: 0.1520
Epoch 313/1000
57/57          0s 4ms/step - loss:
0.0962 - mae: 0.0962 - val_loss: 0.1704 - val_mae: 0.1704
Epoch 314/1000
57/57          0s 4ms/step - loss:
0.0945 - mae: 0.0945 - val_loss: 0.1490 - val_mae: 0.1490
Epoch 315/1000
57/57          0s 4ms/step - loss:
0.0973 - mae: 0.0973 - val_loss: 0.1242 - val_mae: 0.1242
Epoch 316/1000
57/57          0s 4ms/step - loss:
0.0937 - mae: 0.0937 - val_loss: 0.1065 - val_mae: 0.1065
Epoch 317/1000
57/57          0s 4ms/step - loss:
0.0927 - mae: 0.0927 - val_loss: 0.1307 - val_mae: 0.1307
Epoch 318/1000
57/57          0s 4ms/step - loss:
0.0938 - mae: 0.0938 - val_loss: 0.1085 - val_mae: 0.1085
Epoch 319/1000
57/57          0s 4ms/step - loss:
0.0960 - mae: 0.0960 - val_loss: 0.1184 - val_mae: 0.1184
Epoch 320/1000
57/57          0s 4ms/step - loss:
0.0919 - mae: 0.0919 - val_loss: 0.1082 - val_mae: 0.1082
Epoch 321/1000
57/57          0s 4ms/step - loss:
0.0931 - mae: 0.0931 - val_loss: 0.1290 - val_mae: 0.1290
Epoch 322/1000
57/57          0s 4ms/step - loss:
0.0964 - mae: 0.0964 - val_loss: 0.1419 - val_mae: 0.1419
Epoch 323/1000

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57/57          0s 4ms/step - loss:
0.0961 - mae: 0.0961 - val_loss: 0.1355 - val_mae: 0.1355
Epoch 324/1000
57/57          0s 4ms/step - loss:
0.0929 - mae: 0.0929 - val_loss: 0.1096 - val_mae: 0.1096
Epoch 325/1000
57/57          0s 4ms/step - loss:
0.0928 - mae: 0.0928 - val_loss: 0.1089 - val_mae: 0.1089
Epoch 326/1000
57/57          0s 4ms/step - loss:
0.0953 - mae: 0.0953 - val_loss: 0.1153 - val_mae: 0.1153
Epoch 327/1000
57/57          0s 4ms/step - loss:
0.0960 - mae: 0.0960 - val_loss: 0.1102 - val_mae: 0.1102
Epoch 328/1000
57/57          0s 4ms/step - loss:
0.0939 - mae: 0.0939 - val_loss: 0.1081 - val_mae: 0.1081
Epoch 329/1000
57/57          0s 4ms/step - loss:
0.0924 - mae: 0.0924 - val_loss: 0.1083 - val_mae: 0.1083
Epoch 330/1000
57/57          0s 4ms/step - loss:
0.0928 - mae: 0.0928 - val_loss: 0.1387 - val_mae: 0.1387
Epoch 331/1000
57/57          0s 4ms/step - loss:
0.0919 - mae: 0.0919 - val_loss: 0.1095 - val_mae: 0.1095
Epoch 332/1000
57/57          0s 4ms/step - loss:
0.0938 - mae: 0.0938 - val_loss: 0.1485 - val_mae: 0.1485
Epoch 333/1000
57/57          0s 4ms/step - loss:
0.0960 - mae: 0.0960 - val_loss: 0.1108 - val_mae: 0.1108
Epoch 334/1000
57/57          0s 4ms/step - loss:
0.0931 - mae: 0.0931 - val_loss: 0.1157 - val_mae: 0.1157
Epoch 335/1000
57/57          0s 4ms/step - loss:
0.0919 - mae: 0.0919 - val_loss: 0.1192 - val_mae: 0.1192
Epoch 336/1000
57/57          0s 4ms/step - loss:
0.0906 - mae: 0.0906 - val_loss: 0.1097 - val_mae: 0.1097
Epoch 337/1000
57/57          0s 4ms/step - loss:
0.0914 - mae: 0.0914 - val_loss: 0.1371 - val_mae: 0.1371
Epoch 338/1000
57/57          0s 4ms/step - loss:
0.0918 - mae: 0.0918 - val_loss: 0.1081 - val_mae: 0.1081
Epoch 339/1000

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57/57          0s 4ms/step - loss:
0.0918 - mae: 0.0918 - val_loss: 0.1271 - val_mae: 0.1271
Epoch 340/1000
57/57          0s 4ms/step - loss:
0.0883 - mae: 0.0883 - val_loss: 0.1075 - val_mae: 0.1075
Epoch 341/1000
57/57          0s 4ms/step - loss:
0.0953 - mae: 0.0953 - val_loss: 0.1436 - val_mae: 0.1436
Epoch 342/1000
57/57          0s 4ms/step - loss:
0.0936 - mae: 0.0936 - val_loss: 0.1399 - val_mae: 0.1399
Epoch 343/1000
57/57          0s 4ms/step - loss:
0.0904 - mae: 0.0904 - val_loss: 0.1519 - val_mae: 0.1519
Epoch 344/1000
57/57          0s 4ms/step - loss:
0.0935 - mae: 0.0935 - val_loss: 0.1135 - val_mae: 0.1135
Epoch 345/1000
57/57          0s 4ms/step - loss:
0.0898 - mae: 0.0898 - val_loss: 0.1728 - val_mae: 0.1728
Epoch 346/1000
57/57          0s 4ms/step - loss:
0.0933 - mae: 0.0933 - val_loss: 0.1374 - val_mae: 0.1374
Epoch 347/1000
57/57          0s 4ms/step - loss:
0.0928 - mae: 0.0928 - val_loss: 0.1284 - val_mae: 0.1284
Epoch 348/1000
57/57          0s 4ms/step - loss:
0.0891 - mae: 0.0891 - val_loss: 0.1080 - val_mae: 0.1080
Epoch 349/1000
57/57          0s 4ms/step - loss:
0.0904 - mae: 0.0904 - val_loss: 0.1249 - val_mae: 0.1249
Epoch 350/1000
57/57          0s 4ms/step - loss:
0.0928 - mae: 0.0928 - val_loss: 0.1447 - val_mae: 0.1447
Epoch 351/1000
57/57          0s 4ms/step - loss:
0.0918 - mae: 0.0918 - val_loss: 0.1118 - val_mae: 0.1118
Epoch 352/1000
57/57          0s 4ms/step - loss:
0.0885 - mae: 0.0885 - val_loss: 0.1573 - val_mae: 0.1573
Epoch 353/1000
57/57          0s 4ms/step - loss:
0.0934 - mae: 0.0934 - val_loss: 0.1130 - val_mae: 0.1130
Epoch 354/1000
57/57          0s 4ms/step - loss:
0.0898 - mae: 0.0898 - val_loss: 0.1544 - val_mae: 0.1544
Epoch 355/1000

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57/57          0s 4ms/step - loss:
0.0893 - mae: 0.0893 - val_loss: 0.1298 - val_mae: 0.1298
Epoch 356/1000
57/57          0s 4ms/step - loss:
0.0937 - mae: 0.0937 - val_loss: 0.1249 - val_mae: 0.1249
Epoch 357/1000
57/57          0s 4ms/step - loss:
0.0905 - mae: 0.0905 - val_loss: 0.1350 - val_mae: 0.1350
Epoch 358/1000
57/57          0s 4ms/step - loss:
0.0930 - mae: 0.0930 - val_loss: 0.1149 - val_mae: 0.1149
Epoch 359/1000
57/57          0s 4ms/step - loss:
0.0923 - mae: 0.0923 - val_loss: 0.1060 - val_mae: 0.1060
Epoch 360/1000
57/57          0s 4ms/step - loss:
0.0895 - mae: 0.0895 - val_loss: 0.1414 - val_mae: 0.1414
Epoch 361/1000
57/57          0s 4ms/step - loss:
0.0899 - mae: 0.0899 - val_loss: 0.1046 - val_mae: 0.1046
Epoch 362/1000
57/57          0s 4ms/step - loss:
0.0892 - mae: 0.0892 - val_loss: 0.1631 - val_mae: 0.1631
Epoch 363/1000
57/57          0s 4ms/step - loss:
0.0918 - mae: 0.0918 - val_loss: 0.1446 - val_mae: 0.1446
Epoch 364/1000
57/57          0s 4ms/step - loss:
0.0899 - mae: 0.0899 - val_loss: 0.1401 - val_mae: 0.1401
Epoch 365/1000
57/57          0s 4ms/step - loss:
0.0912 - mae: 0.0912 - val_loss: 0.1235 - val_mae: 0.1235
Epoch 366/1000
57/57          0s 4ms/step - loss:
0.0923 - mae: 0.0923 - val_loss: 0.1174 - val_mae: 0.1174
Epoch 367/1000
57/57          0s 4ms/step - loss:
0.0934 - mae: 0.0934 - val_loss: 0.1154 - val_mae: 0.1154
Epoch 368/1000
57/57          0s 4ms/step - loss:
0.0918 - mae: 0.0918 - val_loss: 0.1262 - val_mae: 0.1262
Epoch 369/1000
57/57          0s 4ms/step - loss:
0.0934 - mae: 0.0934 - val_loss: 0.1149 - val_mae: 0.1149
Epoch 370/1000
57/57          0s 4ms/step - loss:
0.0916 - mae: 0.0916 - val_loss: 0.1104 - val_mae: 0.1104
Epoch 371/1000

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57/57 0s 4ms/step - loss:
 0.0894 - mae: 0.0894 - val_loss: 0.1129 - val_mae: 0.1129
 Epoch 372/1000
 57/57 0s 4ms/step - loss:
 0.0874 - mae: 0.0874 - val_loss: 0.1240 - val_mae: 0.1240
 Epoch 373/1000
 57/57 0s 4ms/step - loss:
 0.0866 - mae: 0.0866 - val_loss: 0.1041 - val_mae: 0.1041
 Epoch 374/1000
 57/57 0s 4ms/step - loss:
 0.0899 - mae: 0.0899 - val_loss: 0.1106 - val_mae: 0.1106
 Epoch 375/1000
 57/57 0s 4ms/step - loss:
 0.0928 - mae: 0.0928 - val_loss: 0.1132 - val_mae: 0.1132
 Epoch 376/1000
 57/57 0s 4ms/step - loss:
 0.0897 - mae: 0.0897 - val_loss: 0.1060 - val_mae: 0.1060
 Epoch 377/1000
 57/57 0s 4ms/step - loss:
 0.0900 - mae: 0.0900 - val_loss: 0.1300 - val_mae: 0.1300
 Epoch 378/1000
 57/57 0s 4ms/step - loss:
 0.0873 - mae: 0.0873 - val_loss: 0.1067 - val_mae: 0.1067
 Epoch 379/1000
 57/57 0s 4ms/step - loss:
 0.0894 - mae: 0.0894 - val_loss: 0.1395 - val_mae: 0.1395
 Epoch 380/1000
 57/57 0s 4ms/step - loss:
 0.0915 - mae: 0.0915 - val_loss: 0.1230 - val_mae: 0.1230
 Epoch 381/1000
 57/57 0s 4ms/step - loss:
 0.0873 - mae: 0.0873 - val_loss: 0.1104 - val_mae: 0.1104
 Epoch 382/1000
 57/57 0s 4ms/step - loss:
 0.0875 - mae: 0.0875 - val_loss: 0.1325 - val_mae: 0.1325
 Epoch 383/1000
 57/57 0s 4ms/step - loss:
 0.0910 - mae: 0.0910 - val_loss: 0.1143 - val_mae: 0.1143
 Epoch 384/1000
 57/57 0s 4ms/step - loss:
 0.0898 - mae: 0.0898 - val_loss: 0.1292 - val_mae: 0.1292
 Epoch 385/1000
 57/57 0s 4ms/step - loss:
 0.0865 - mae: 0.0865 - val_loss: 0.1012 - val_mae: 0.1012
 Epoch 386/1000
 57/57 0s 4ms/step - loss:
 0.0896 - mae: 0.0896 - val_loss: 0.1280 - val_mae: 0.1280
 Epoch 387/1000

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57/57          0s 4ms/step - loss:
0.0931 - mae: 0.0931 - val_loss: 0.1412 - val_mae: 0.1412
Epoch 388/1000
57/57          0s 4ms/step - loss:
0.0897 - mae: 0.0897 - val_loss: 0.1044 - val_mae: 0.1044
Epoch 389/1000
57/57          0s 4ms/step - loss:
0.0881 - mae: 0.0881 - val_loss: 0.1201 - val_mae: 0.1201
Epoch 390/1000
57/57          0s 4ms/step - loss:
0.0936 - mae: 0.0936 - val_loss: 0.1370 - val_mae: 0.1370
Epoch 391/1000
57/57          0s 4ms/step - loss:
0.0887 - mae: 0.0887 - val_loss: 0.1655 - val_mae: 0.1655
Epoch 392/1000
57/57          0s 4ms/step - loss:
0.0872 - mae: 0.0872 - val_loss: 0.1346 - val_mae: 0.1346
Epoch 393/1000
57/57          0s 4ms/step - loss:
0.0885 - mae: 0.0885 - val_loss: 0.1217 - val_mae: 0.1217
Epoch 394/1000
57/57          0s 4ms/step - loss:
0.0904 - mae: 0.0904 - val_loss: 0.1018 - val_mae: 0.1018
Epoch 395/1000
57/57          0s 4ms/step - loss:
0.0853 - mae: 0.0853 - val_loss: 0.1067 - val_mae: 0.1067
Epoch 396/1000
57/57          0s 4ms/step - loss:
0.0914 - mae: 0.0914 - val_loss: 0.1120 - val_mae: 0.1120
Epoch 397/1000
57/57          0s 4ms/step - loss:
0.0882 - mae: 0.0882 - val_loss: 0.1346 - val_mae: 0.1346
Epoch 398/1000
57/57          0s 4ms/step - loss:
0.0880 - mae: 0.0880 - val_loss: 0.1110 - val_mae: 0.1110
Epoch 399/1000
57/57          0s 4ms/step - loss:
0.0879 - mae: 0.0879 - val_loss: 0.1164 - val_mae: 0.1164
Epoch 400/1000
57/57          0s 4ms/step - loss:
0.0902 - mae: 0.0902 - val_loss: 0.1414 - val_mae: 0.1414
Epoch 401/1000
57/57          0s 4ms/step - loss:
0.0901 - mae: 0.0901 - val_loss: 0.1234 - val_mae: 0.1234
Epoch 402/1000
57/57          0s 4ms/step - loss:
0.0868 - mae: 0.0868 - val_loss: 0.1341 - val_mae: 0.1341
Epoch 403/1000

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57/57          0s 4ms/step - loss:
0.0866 - mae: 0.0866 - val_loss: 0.1419 - val_mae: 0.1419
Epoch 404/1000
57/57          0s 4ms/step - loss:
0.0924 - mae: 0.0924 - val_loss: 0.1137 - val_mae: 0.1137
Epoch 405/1000
57/57          0s 4ms/step - loss:
0.0867 - mae: 0.0867 - val_loss: 0.1072 - val_mae: 0.1072
Epoch 406/1000
57/57          0s 4ms/step - loss:
0.0876 - mae: 0.0876 - val_loss: 0.1376 - val_mae: 0.1376
Epoch 407/1000
57/57          0s 4ms/step - loss:
0.0877 - mae: 0.0877 - val_loss: 0.1146 - val_mae: 0.1146
Epoch 408/1000
57/57          0s 4ms/step - loss:
0.0892 - mae: 0.0892 - val_loss: 0.1211 - val_mae: 0.1211
Epoch 409/1000
57/57          0s 4ms/step - loss:
0.0876 - mae: 0.0876 - val_loss: 0.1285 - val_mae: 0.1285
Epoch 410/1000
57/57          0s 4ms/step - loss:
0.0862 - mae: 0.0862 - val_loss: 0.1132 - val_mae: 0.1132
Epoch 411/1000
57/57          0s 4ms/step - loss:
0.0868 - mae: 0.0868 - val_loss: 0.1150 - val_mae: 0.1150
Epoch 412/1000
57/57          0s 4ms/step - loss:
0.0869 - mae: 0.0869 - val_loss: 0.1182 - val_mae: 0.1182
Epoch 413/1000
57/57          0s 4ms/step - loss:
0.0846 - mae: 0.0846 - val_loss: 0.1177 - val_mae: 0.1177
Epoch 414/1000
57/57          0s 4ms/step - loss:
0.0895 - mae: 0.0895 - val_loss: 0.1233 - val_mae: 0.1233
Epoch 415/1000
57/57          0s 4ms/step - loss:
0.0884 - mae: 0.0884 - val_loss: 0.1359 - val_mae: 0.1359
Epoch 416/1000
57/57          0s 4ms/step - loss:
0.0845 - mae: 0.0845 - val_loss: 0.1265 - val_mae: 0.1265
Epoch 417/1000
57/57          0s 4ms/step - loss:
0.0886 - mae: 0.0886 - val_loss: 0.1377 - val_mae: 0.1377
Epoch 418/1000
57/57          0s 4ms/step - loss:
0.0898 - mae: 0.0898 - val_loss: 0.1176 - val_mae: 0.1176
Epoch 419/1000

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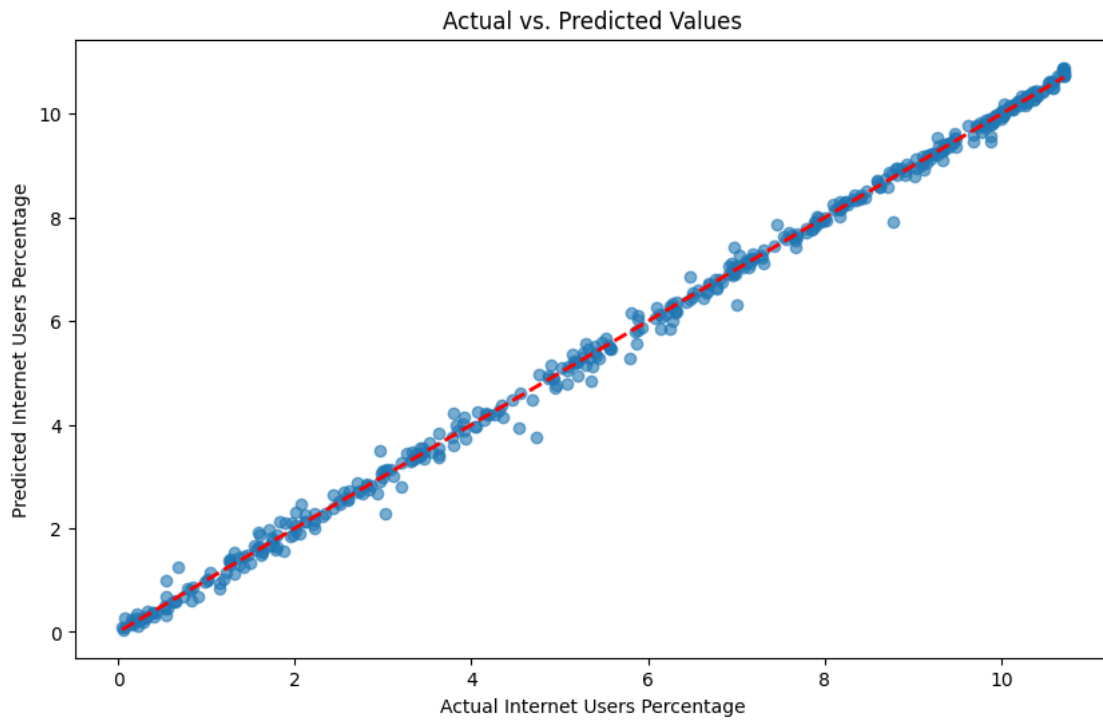
```

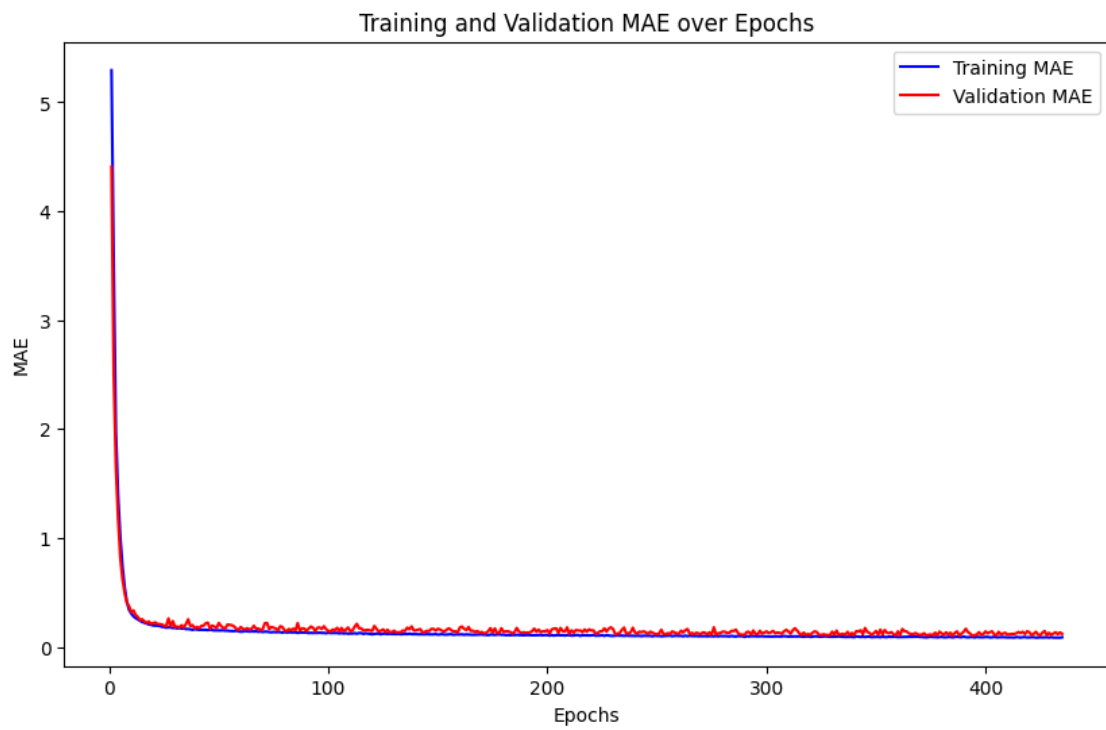
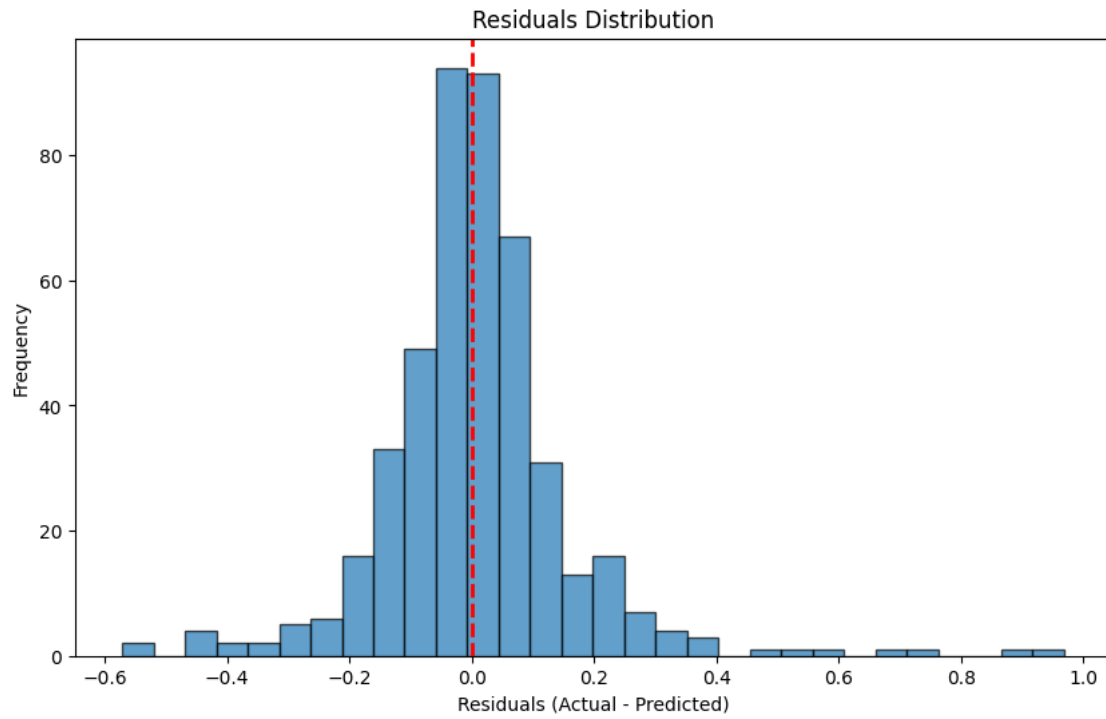
57/57          0s 4ms/step - loss:
0.0857 - mae: 0.0857 - val_loss: 0.1424 - val_mae: 0.1424
Epoch 420/1000
57/57          0s 4ms/step - loss:
0.0863 - mae: 0.0863 - val_loss: 0.1348 - val_mae: 0.1348
Epoch 421/1000
57/57          0s 4ms/step - loss:
0.0868 - mae: 0.0868 - val_loss: 0.1069 - val_mae: 0.1069
Epoch 422/1000
57/57          0s 4ms/step - loss:
0.0884 - mae: 0.0884 - val_loss: 0.1199 - val_mae: 0.1199
Epoch 423/1000
57/57          0s 4ms/step - loss:
0.0847 - mae: 0.0847 - val_loss: 0.1376 - val_mae: 0.1376
Epoch 424/1000
57/57          0s 4ms/step - loss:
0.0850 - mae: 0.0850 - val_loss: 0.1130 - val_mae: 0.1130
Epoch 425/1000
57/57          0s 4ms/step - loss:
0.0875 - mae: 0.0875 - val_loss: 0.1057 - val_mae: 0.1057
Epoch 426/1000
57/57          0s 4ms/step - loss:
0.0858 - mae: 0.0858 - val_loss: 0.1302 - val_mae: 0.1302
Epoch 427/1000
57/57          0s 4ms/step - loss:
0.0864 - mae: 0.0864 - val_loss: 0.1452 - val_mae: 0.1452
Epoch 428/1000
57/57          0s 4ms/step - loss:
0.0876 - mae: 0.0876 - val_loss: 0.1063 - val_mae: 0.1063
Epoch 429/1000
57/57          0s 4ms/step - loss:
0.0845 - mae: 0.0845 - val_loss: 0.1283 - val_mae: 0.1283
Epoch 430/1000
57/57          0s 4ms/step - loss:
0.0891 - mae: 0.0891 - val_loss: 0.1100 - val_mae: 0.1100
Epoch 431/1000
57/57          0s 4ms/step - loss:
0.0846 - mae: 0.0846 - val_loss: 0.1222 - val_mae: 0.1222
Epoch 432/1000
57/57          0s 4ms/step - loss:
0.0842 - mae: 0.0842 - val_loss: 0.1338 - val_mae: 0.1338
Epoch 433/1000
57/57          0s 4ms/step - loss:
0.0849 - mae: 0.0849 - val_loss: 0.1156 - val_mae: 0.1156
Epoch 434/1000
57/57          0s 4ms/step - loss:
0.0841 - mae: 0.0841 - val_loss: 0.1317 - val_mae: 0.1317
Epoch 435/1000

```


57/57 0s 4ms/step - loss:
0.0878 - mae: 0.0878 - val_loss: 0.1187 - val_mae: 0.1187
Epoch 435: early stopping
Restoring model weights from the end of the best epoch: 385.

Final Model Loss on Test Data: 0.1012
Final Model MAE on Test Data: 0.1012
15/15 0s 6ms/step





[]: